LEARNING MODULE

26

Time Series II: Volatility Models

Tomorrow's variance depends on today's surprise: forecast it with EWMA and GARCH, measure it from intraday TAIEX data, and judge the forecasts against that noisy measurement.

VOLUME IV · Statistical Inference and Econometrics

BUILDS ON LM3 · LM5 · LM13 · LM14 · LM15 · LM16 · LM17 · LM18 · LM19 · LM20 · LM21 · LM23 · LM25

LEADS TO LM30 · LM31 · LM32 · LM34 · LM39

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** explain volatility clustering as dependence between the sizes of returns, and test for it with Engle's ARCH test
☐	**2** compute an EWMA variance by hand, with its weights, effective window and RiskMetrics' choice $\lambda = 0.94$
☐	**3** state the GARCH(1,1) model with its conditions, and derive its unconditional variance, its multi-step forecasts and the half-life of a shock
☐	**4** fit GARCH(1,1) by maximum likelihood with normal and Student t innovations, compute the log-likelihood by hand, and check the fit with standardized residuals
☐	**5** compute realized variance from intraday data, read a volatility signature plot, and state the hypotheses (microstructure noise, stale prices) that could explain how it moves with the sampling interval
☐	**6** evaluate variance forecasts out of sample with MSE and QLIKE against a noisy realized-variance target, and with Mincer–Zarnowitz regressions
☐	**7** run these tools in Python with arch and statsmodels on NumPy arrays, matching each package result to a calculation by hand

LEARNING MODULE OVERVIEW

- TR23's squared returns depend on yesterday's: the ARCH test with five lags gives 696.3, which none of 20,000 shuffles of the days reaches.
- EWMA with RiskMetrics' $\lambda = 0.94$ has an effective window of 16.7 days. After 7 April 2025 it moved the next day's forecast from 1.5605% to 2.9215%.
- GARCH(1,1) adds a long-run level. On TR23, $\alpha + \beta = 0.98443$, so a shock's effect on the variance forecast halves in 44.2 sessions, toward 19.83% a year; mapped from $\alpha + \beta$, the interval for the half-life runs from 26 to 152 sessions.
- The normal GARCH fit leaves standardized residuals with excess kurtosis 2.57, against 5.428 for the returns. With t innovations the log-likelihood rises by 183.35 at $\hat{\nu} = 5.54$.
- Realized variance from 5-second data depends on the sampling interval: in 2025 it was 24.3 times the 5-minute value, while the TX future's barely moved from 1 to 30 minutes.
- The daily target is the opening return squared plus the 15-minute realized variance of 09:30–13:30. On average two thirds of it is that single squared opening return.
- Out of sample over 4,609 days, GARCH has the lowest QLIKE, 0.5032; EWMA's 0.5153 is 1.83 Newey–West standard errors worse, and yesterday's realized variance is worst.

1 INTRODUCTION

LM13 to LM18 found that large moves come in clusters, and LM19 measured it: TR23's squared log returns correlate 0.194 with the previous day's. This module forecasts tomorrow's variance from that dependence, measures a day's variance from intraday index data, and judges the forecasts against that noisy measurement.

MOTIVATING CASE **The day after the worst day**

On Wednesday 2 April 2025, the last session before the Qingming holiday, a GARCH(1,1) model refitted on the previous five years forecast a daily standard deviation of 1.72%. That is 27.0% a year. An exponentially weighted moving average (EWMA) said 1.56%. On Monday 7 April the TAIEX total-return index fell 9.70%, a log return of −10.20%: 5.9 of GARCH's standard deviations, and TR23's largest fall (LM15).

On the evening of 7 April the forecasts for 8 April diverged: GARCH 3.60%, EWMA 2.92%, and 7 April's own realized volatility, read as a forecast, 10.09%. The day's realized measure, $\sqrt{V_t}$ from intraday TAIEX data, then put 8 April at 3.39% (Exhibit 1). Which forecast should a risk manager trust, how fast should it fall back, and how can a forecast be judged when a day's true variance is never observed?

Exhibit 1: Forecasts of the daily standard deviation around 7 April 2025

Made on → for	GARCH	EWMA 0.94	60-day	Yesterday's V	Realized $\sqrt{V_t}$	r_t
2 April → 7 April	1.7222	1.5605	1.3226	0.7004	10.0859	−10.2029
7 April → 8 April	3.6031	2.9215	1.8665	10.0859	3.3873	−4.0990

Note: % a day; forecasts out of sample, as in Section 7. The target V_t is the opening return squared plus the 15-minute realized variance of 09:30–13:30; on 7 April 99.99% of it came before 09:30. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026, `TaiwanVariousIndicators5Seconds` (TWSE's TAIEX), retrieved 19 October 2026, and `TaiwanStockPrice` (the TAIEX closes before each opening return), retrieved 11 and 18 October 2026. Computations: `data/lm26_numbers.py`.

Section 2 tests for clustering, Sections 3 to 5 build the forecasts, Section 6 measures a day's variance, and Section 7 judges them.

> **NOTE** **Notation for this module**
>
> Daily data are TR23: $T = 5{,}849$ log returns $r_t = \ln(1 + R_t)$ from 3 January 2003 to 8 October 2026, $A = 246.12$ (LM17). Models use r_t in percent, so variances are in %² a day. Intraday data are the TAIEX price index every 5 to 60 seconds, 5,351 sessions from 3 January 2005. Moments divide by T. Here α and β are GARCH coefficients, so a significance level is written "the 5% level".
>
> Local: V_t, $r_{\mathrm{o},t}$ the daily target and the opening return; $\mathrm{RV}_t(h)$ with $n = 14{,}400$ s$/h$ returns indexed by i (no generic function h); x_i, x_i^{e}, u_i, σ_{e}, σ_u the observed and efficient log index and the noise (no probability level u); J intervals of length h in one of Jh; K the ARCH lags; a, b_k and a, b auxiliary regression coefficients (no bets); m a horizon in sessions; LM25's η_t in Deep Dive 1; intraday $\hat{\rho}_1$: uncentred, pooled over a period's within-day pairs of 09:30–13:30 returns (a single day's own $\hat{\rho}_1$ is centred at that day's mean). Counts: t days, i intraday returns, j and k lags.

> **NOTE** **Research hygiene**
>
> The ARCH tests and the full-sample fits are **in-sample**. Section 7's forecasts are **out-of-sample**, with the 60-day window and $\lambda = 0.94$ fixed in advance. Outside Deep Dive 3, intraday statistics, R-squared values and sample moments are **descriptive**, without standard errors (LM20). Simulations, the shuffles, the widget and the noise model are **models**; other windows and single years are **stability checks**.
>
> Tests on these data number 25 (LM21's count). They are five ARCH (two by permutation), two Ljung–Box (one is LM25's, recomputed), ten loss differences, six Mincer–Zarnowitz slopes, V_t against r_t^2 and Deep Dive 3's extremes. Nothing is tested on later data (LM34).

2 VOLATILITY CLUSTERING AND THE ARCH TEST

- [ ] explain volatility clustering as dependence between the sizes of returns, and test for it with Engle's ARCH test

The evidence so far concerns the sizes of returns, not their signs. After a fall of 3% or more TR23 fell again 12.04% of the time, against 1.85% overall (LM13). Next-day log returns correlate only 0.0393, but their sizes correlate 0.161 (LM19). LM25's Ljung–Box statistic for $|r_t|$, (25.5), is 1,820 on ten lags.

Volatility clustering is this dependence between sizes. A model for it describes the **conditional variance** σ_t^2, the variance of day t given the returns before it:

$$r_t = g + \varepsilon_t, \qquad \varepsilon_t = \sigma_t \xi_t.$$

Here $\sigma_t > 0$ depends only on earlier returns, and the ξ_t are iid with mean 0 and variance 1, each independent of the past. By the tower property (16.5), $\mathbb{E}[\varepsilon_t \varepsilon_{t-k}] = \mathbb{E}\big[\mathbb{E}[\varepsilon_t \mid \text{past}]\, \varepsilon_{t-k}\big] = 0$ for $k \geq 1$. So the innovations are uncorrelated, yet ε_t^2 and ε_{t-1}^2 correlate whenever σ_t grows with $|\varepsilon_{t-1}|$ and fourth moments are finite.

RESULT **The ARCH test**

Let $\hat{\varepsilon}_t = r_t - \bar{r}$. Regress $\hat{\varepsilon}_t^2$ by OLS, (23.1), on a constant and $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-K}^2$, with coefficients $a, b_1, \dots, b_K$, over the $T - K$ days that have K lags. Under H_0 the ε_t are iid with a finite fourth moment, and then, for large T, the statistic

$$(T - K) \times \text{R-squared} \quad \text{is approximately } \chi^2_K. \tag{26.1}$$

The test uses the upper tail. Stated without proof (Engle 1982).

EXAMPLE 1

The ARCH test on TR23

Test TR23's log returns, in percent, with $K = 1$ and with $K = 5$.

SOLUTION

With $K = 1$ the intercept is $a = 1.1779$ and the slope $b_1 = 0.196$. The slope is close to LM19's $\hat{\rho}_1$ of r_t^2, 0.194, because regressing a stationary series on its lag gives about its lag-1 autocorrelation. R-squared is 0.03844, so the statistic is $5{,}848 \times 0.03844 = 224.8$, with a p-value of 8.2×10^{-51} from χ^2_1.

With $K = 5$ the statistic is 696.3, far above the 5% point 11.07 of χ^2_5; its p-value is below 10^{-100}. These χ^2 p-values are nominal: (26.1) needs a finite fourth moment, which LM15's fit ($\nu = 3.065$) lacks (LM19). The five slopes sum to 0.529.

Experiment 3 shuffles TR23's days 20,000 times (`default_rng([3, 20261011])`), as LM21's experiment 8 shuffled dates; no shuffle reaches 224.8 or 696.3. A constant variance is rejected beyond doubt.

PITFALL **The ARCH test names no model**

Any change in variance makes the test reject. After a calm decade and a turbulent one, squared residuals correlate with their lags, because a day and its lag usually share a decade. A rejection says only that the variance is not constant.

導讀 **大小會群聚，方向不會**

報酬的正負號幾乎無法預測，報酬的大小卻會成群出現，這就是波動群聚現象（volatility clustering）。模型描述的是條件變異數（conditional variance）：在已知昨天以前所有報酬的條件下，今天報酬的變異數。新息（innovation）的條件期望值（conditional expectation）是零，所以互不相關（uncorrelated）；但只要今天的變異數取決於昨天的意外有多大，它們的平方就會相關。Engle 的自我迴歸條件變異數非齊一檢定（ARCH test）把殘差（residual）平方對自己前幾期的值（lagged values）做迴歸（regression），對立假設（alternative hypothesis）是自我迴歸條件變異數非齊一模型（ARCH model）；在虛無假設（null hypothesis）下，若四階動差（fourth moment）有限，統計量（statistic）近似卡方分布（chi-square distribution）。檢定拒絕只代表變異數不是常數，並沒有指出是哪一種模型。

KNOWLEDGE CHECK 1

The ARCH regression with $K = 1$ has $a = 1.1779$ and $b_1 = 0.1960$. What variance does it predict for tomorrow after a surprise of 3% today, and after one of 0.5%?

ANSWER

After 3%: $1.1779 + 0.1960 \times 9 = 2.94\ \%^2$, a standard deviation of 1.72%. After 0.5%: $1.1779 + 0.1960 \times 0.25 = 1.23\ \%^2$, or 1.11%. Today's size moves tomorrow's spread, as LM16 described.

3 THE EXPONENTIALLY WEIGHTED MOVING AVERAGE

- ☐ compute an EWMA variance by hand, with its weights, effective window and RiskMetrics' choice $\lambda = 0.94$

The simplest variance forecast is the mean of the last 60 squared returns, in which a crash counts fully for 60 days and then drops out at once. An exponentially weighted average lets old days fade instead.

DEFINITION **EWMA variance**

With a decay $0 < \lambda < 1$ and returns of mean zero, the **exponentially weighted moving average** forecast of tomorrow's variance is

$$\begin{aligned} \sigma^2_{t+1|t} &= \lambda\, \sigma^2_{t|t-1} + (1-\lambda)\, r_t^2 \\ &= (1-\lambda) \sum_{j \geq 0} \lambda^j r^2_{t-j}. \end{aligned} \tag{26.2}$$

The second form follows by substituting the recursion into itself (Problem 7). The weights $(1-\lambda)\lambda^j$ sum to 1 and halve every $\ln \tfrac{1}{2} / \ln \lambda$ days. The **effective window** $1/(1-\lambda)$ is the number of equal weights that would give today the same weight.

RiskMetrics (1996, §5.3.2.2) fixed $\lambda = 0.94$ for daily data. That is a weighted average, over 480 series, of the λ that minimizes each series' root mean squared error of variance forecasts, weighted by forecast accuracy. Its §5.3.1.2 counts the effective days as those that leave out a tolerance level of the weight: at 1%, $\ln 0.01 / \ln 0.94 = 74.4$, which its Table 5.7 lists as 74 days. RiskMetrics sets the mean return to zero (its Table 5.9), as in (26.2).

EXAMPLE 2

EWMA across 7 April 2025

The EWMA forecast for 7 April was $2.4351\ \%^2$, or 1.5605%. Update it with 7 April's log return, −10.2029%, and describe the weights at $\lambda = 0.94$.

SOLUTION

In %², $r_t^2 = 104.0999$. By (26.2),

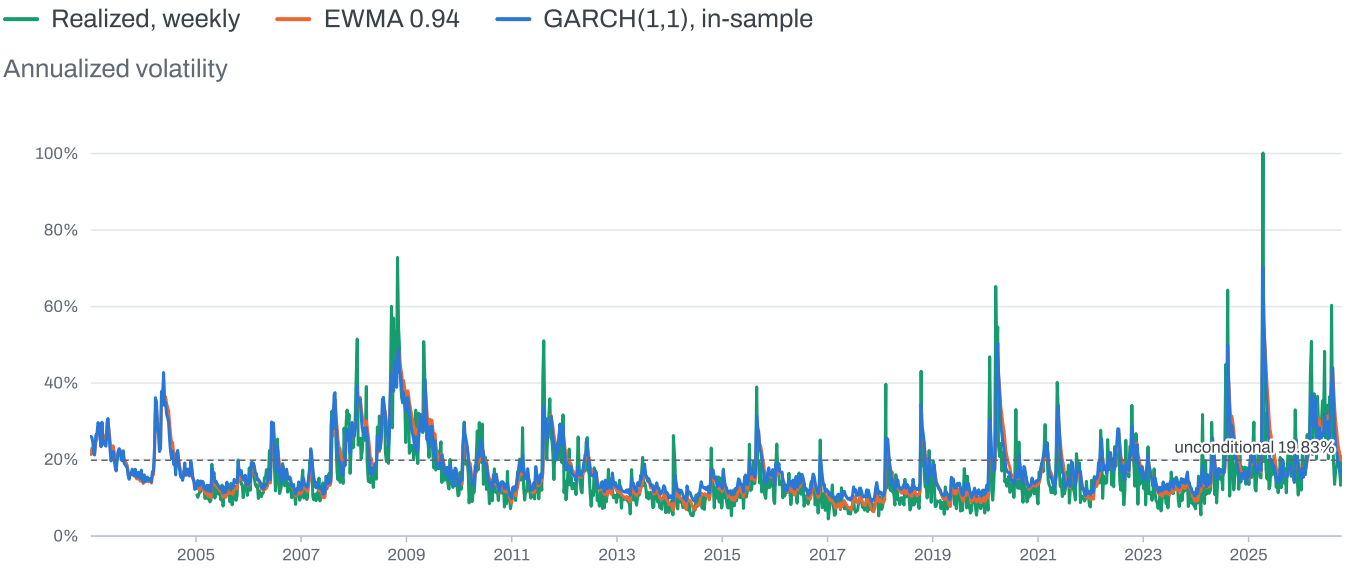
$$\begin{aligned}\sigma_{t+1|t}^2 &= 0.94 \times 2.4351 + 0.06 \times 104.0999 \\ &= 2.2890 + 6.2460,\end{aligned}$$

so the forecast for 8 April is $\sqrt{8.5350} = 2.9215\%$, nearly double.

The effective window is 16.7 days, the weights halve every 11.2 days, and the last 75 days carry 99.0% of the weight. The recursion started at 1.84 %², the mean of r_t^2 over the first 246 returns. In the forecast for 2 January 2008, Section 7's first test day, that start carries a weight of $0.94^{1,240} = 5 \times 10^{-34}$.

Exhibit 2 shows the EWMA forecast by week beside Section 6's realized volatility: it rises within days of a crash and decays over weeks.

Exhibit 2: TAIEX volatility by week: GARCH, EWMA and realized



Note: annualized, at each ISO week's last session (1,224 weeks). GARCH: $\sqrt{A}\,\sigma_t$ from the full-sample normal fit of Exhibit 3, in-sample. EWMA: $\sqrt{A}\,\sigma_{t|t-1}$, $\lambda = 0.94$. Realized: $\sqrt{A \times \text{the week's mean } V_t}$, from 2005 (1,120 weeks). Dashed: GARCH's unconditional 19.83%. All three peak in the week ending 11 April 2025, at 70.41%, 60.11% and 100.16%. Source: as Exhibit 1.

EXAMPLE 3

Choosing λ in hindsight

Section 7 scores forecasts by QLIKE, (26.8), against V_t on 4,609 days. Which λ would have scored best?

SOLUTION

On a grid from 0.800 to 0.995 in steps of 0.005, QLIKE is 0.5153 at 0.94, 0.5436 at 0.97 and lowest, 0.5087, at 0.905. GARCH scores 0.5032. Choosing 0.905 uses the test days, so that choice is in-sample.

PITFALL **A decay chosen on the test days**

The best λ of Example 3 beats 0.94 on the days it was chosen on, by construction. Only a value fixed in advance, as RiskMetrics' 0.94 was, gives an out-of-sample score.

導讀 **記憶逐漸淡去，而不是一刀切**

60 天移動平均（moving average）把每一天都當成同等重要，滿 60 天就一次丟掉，所以預測值（forecast）會突然跳動。指數加權移動平均模型（EWMA）讓每一天的權重（weight）以 λ 的比例逐日遞減：昨天的平方報酬占 $1-\lambda$，越早的日子越不重要，但不會一下子歸零。RiskMetrics 對日資料用 $\lambda = 0.94$，有效視窗（effective window）約 16.7 天，權重每 11.2 天減半。崩盤隔天，波動率（volatility）預測立刻上升，之後慢慢回落；回落到哪裡，取決於最近的平靜程度，因為 EWMA 沒有長期水準。

KNOWLEDGE CHECK 2

With $\lambda = 0.97$, what is the effective window, and in how many days do the weights halve?

ANSWER

The effective window is $1/0.03 = 33.3$ days, and the weights halve in $\ln\frac{1}{2}/\ln 0.97 = 22.76$ days.

4 GARCH(1,1): MODEL, FORECASTS AND HALF-LIFE

- ☐ state the GARCH(1,1) model with its conditions, and derive its unconditional variance, its multi-step forecasts and the half-life of a shock

EWMA has no long-run level, so its forecast for every later day equals tomorrow's (Deep Dive 1). GARCH, generalized ARCH (Bollerslev 1986), adds a constant that pulls the variance back toward a level.

DEFINITION **GARCH(1,1)**

Let $r_t = g + \varepsilon_t$ and $\varepsilon_t = \sigma_t \xi_t$, with

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2. \tag{26.3}$$

The model's conditions are:

1. the ξ_t are iid with mean 0 and variance 1, each independent of the returns before day t;
2. a positive constant, $\omega > 0$;
3. non-negative coefficients, $\alpha \geq 0$ and $\beta \geq 0$;
4. a persistence below 1, $\alpha + \beta < 1$.

Tomorrow's variance is a constant plus a share of today's squared surprise and a share of today's variance. The ξ_t may be normal, or Student t scaled to variance 1 by (15.11).

Given the returns up to day t, σ_{t+1}^2 is known, so $\sigma_{t+1|t}^2 = \sigma_{t+1}^2$. For $j \geq 1$, condition 1 and the tower property (16.5) give $\mathbb{E}[\varepsilon_{t+j}^2 \mid \text{day } t] = \sigma_{t+j|t}^2$. Taking that expectation of (26.3) for day $t + j + 1$ gives

$$\sigma_{t+j+1|t}^2 = \omega + (\alpha + \beta)\, \sigma_{t+j|t}^2.$$

A stationary model has a constant unconditional variance $\sigma^2 = \mathbb{E}\varepsilon_t^2$, and the same step on unconditional expectations gives $\sigma^2 = \omega + (\alpha + \beta)\sigma^2$. Subtracting, each step multiplies the forecast's distance from σ^2 by $\alpha + \beta$.

RESULT **Unconditional variance, forecasts and half-life**

Under the conditions of (26.3), with ε_t stationary,

$$\begin{aligned} \sigma^2 &= \frac{\omega}{1 - \alpha - \beta}, \\ \sigma_{t+m|t}^2 &= \sigma^2 + (\alpha + \beta)^{m-1}\big(\sigma_{t+1|t}^2 - \sigma^2\big), \\ \text{half-life} &= \frac{\ln \frac{1}{2}}{\ln(\alpha + \beta)}. \end{aligned} \tag{26.4}$$

The half-life is (25.2)'s: the number of sessions in which a shock's effect on the forecast halves. That a stationary solution exists when $\alpha + \beta < 1$ is stated without proof (Tsay 2010, ch. 3).

DEEP DIVE **Deep Dive 1: the GARCH variance unrolled** optional · click to expand

Here η_t is LM25's white noise. Apply (26.3) k times:

$$\sigma_t^2 = \omega \sum_{j=0}^{k-1} \beta^j + \alpha \sum_{j=0}^{k-1} \beta^j \varepsilon_{t-1-j}^2 + \beta^k \sigma_{t-k}^2.$$

For $\beta < 1$ and bounded $\mathbb{E}\sigma_{t-k}^2$ the last term vanishes as $k \to \infty$. So $\sigma_t^2 = \omega/(1-\beta) + \alpha \sum_{j\geq 0} \beta^j \varepsilon_{t-1-j}^2$: past surprises weighted with decay β, plus a floor. EWMA is the case $\omega = 0$, $\alpha = 1-\lambda$, $\beta = \lambda$, which breaks conditions 2 and 4. With $\omega = 0$ and $\alpha + \beta = 1$ the forecast recursion gives $\sigma_{t+m|t}^2 = \sigma_{t+1|t}^2$ for every m; with $\omega > 0$ it rises by ω a step.

Next, let $\eta_t := \varepsilon_t^2 - \sigma_t^2$, which has mean 0 given the past and is white noise when $\mathbb{E}\varepsilon_t^4 < \infty$. Replacing σ_t^2 and σ_{t-1}^2 in (26.3) by $\varepsilon_t^2 - \eta_t$ and $\varepsilon_{t-1}^2 - \eta_{t-1}$ gives

$$\varepsilon_t^2 = \omega + (\alpha + \beta)\, \varepsilon_{t-1}^2 + \eta_t - \beta\, \eta_{t-1},$$

an ARMA(1,1) in the squares. For $k \geq 2$, η_t and η_{t-1} are uncorrelated with ε_{t-k}^2. So from lag 2 on, each autocovariance of ε_t^2 is $\alpha + \beta$ times the one before, like (25.1)'s AR(1).

EXAMPLE 4

How fast the forecast falls back

Take the full-sample normal fit of Exhibit 3, with $\alpha + \beta = 0.98443$ and $\sigma^2 = 1.5971\ \%^2$. On the evening of 7 April 2025 it forecast $\sigma_{t+1|t} = 3.517\%$ ($12.3695\ \%^2$). Trace the forecasts.

SOLUTION

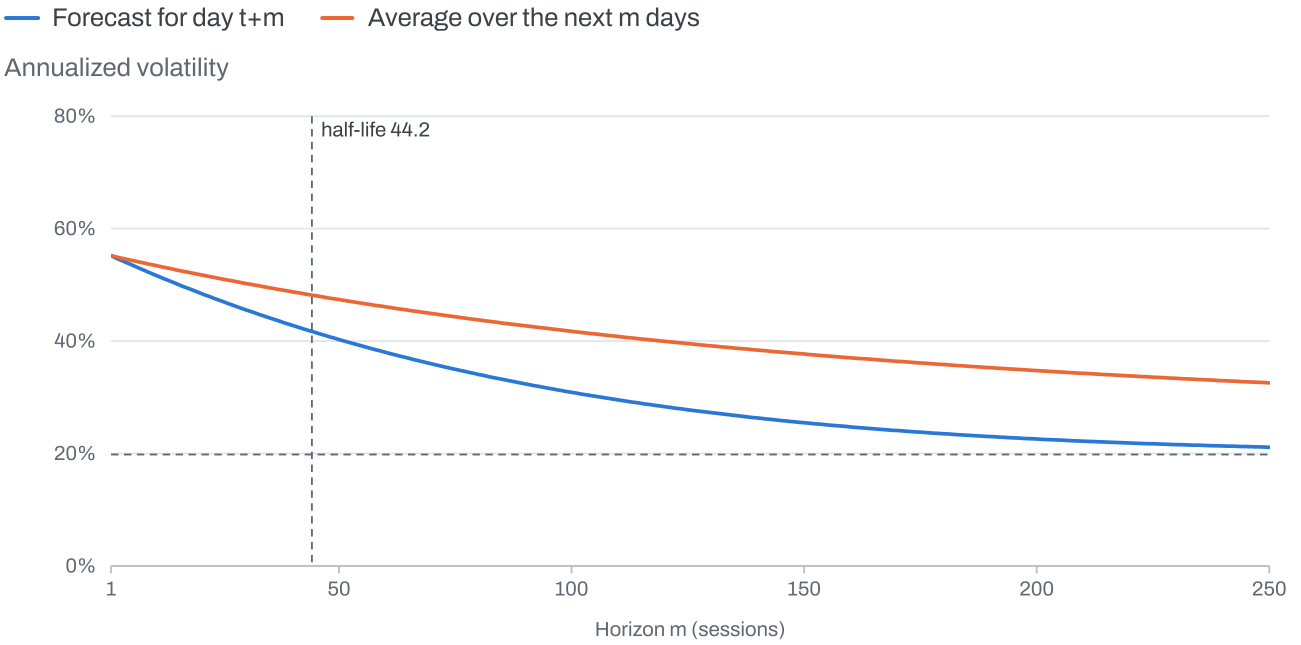
By (26.4) the forecasts fall to 3.42%, 3.10%, 2.66% and 1.35% at m = 5, 20, 44 and 250 sessions, toward $\sigma = 1.264\%$ a day. The half-life is $\ln \frac{1}{2} / \ln 0.98443 = 44.2$ sessions. The average forecast variance over the next 20 days is a volatility of 51.8% a year, and over 250 days 32.6%.

On 8 October 2026, the last day, the forecast was 1.092% (17.13% a year), below 19.83%, so it rises toward that level. arch's `forecast` gives the same values to 10^{-12}.

Starting from Example 4, the widget draws (26.4) for any $\alpha + \beta < 1$ at the fit's unconditional variance. LM31 scales a position by such a forecast, and LM32 turns it into a value-at-risk. In continuous time a variance that moves at random is LM39's stochastic volatility.

How a variance forecast returns to its mean

INTERACTIVE



α + β	Half-life (sessions)	Unconditional volatility, a year
0.98443	**44.2**	**19.83%**
Forecast for tomorrow, a year	**Average over the next 20 days**	**Average over the next 250 days**
55.18%	**51.83%**	**32.57%**

Model: GARCH(1,1) forecasts by (26.4), starting from the next day's forecast of the full-sample normal fit to TR23 (3.517% a day after 7 April 2025, 1.092% after 8 October 2026). The unconditional variance stays at 1.5971 %² a day (19.83% a year, dashed), so ω = 1.5971 × (1 − α − β). When α + β ≥ 1 there is no unconditional variance: ω is set to 0, the forecast stays flat (EWMA) or grows. Values above 80% a year are not drawn.

PITFALL $\sqrt{A}$ **times a one-day forecast is not a year's forecast**

Annualizing tomorrow's forecast by $\sqrt{A}$ assumes that every later day has the same variance. On the evening of 7 April 2025 that gave 55.18% a year, while the average forecast over the next 250 sessions was 32.57%. Average the forecasts over the horizon instead, as the widget does.

導讀 **衝擊會衰退，回到長期水準**

廣義自我迴歸條件變異數非齊一模型（GARCH）說：明天的變異數（variance）等於一個常數，加上今天意外的平方的一部分，再加上今天變異數的一部分。只要 $\alpha + \beta < 1$，就有一個長期的無條件變異數（unconditional variance），預測值（forecast）與它的差距每天乘上 $\alpha + \beta$。差距減半所需的天數就是半衰期（half-life）。加權股價報酬指數的 $\alpha + \beta = 0.98443$，半衰期約 44 個交易日。把明天的預測乘上 $\sqrt{A}$ 當成年化波動率（annualized volatility），等於假設往後每天都一樣，在崩盤之後會高估很多。

KNOWLEDGE CHECK 3

What half-life does $\alpha + \beta = 0.98$ give? What does TR23's t fit, $\alpha + \beta = 0.99098$, give?

ANSWER

By (26.4), $\ln \frac{1}{2} / \ln 0.98 = 34.3$ sessions, and $\ln \frac{1}{2} / \ln 0.99098 = 76.5$ sessions.

5 FITTING GARCH BY MAXIMUM LIKELIHOOD

- [ ] fit GARCH(1,1) by maximum likelihood with normal and Student t innovations, compute the log-likelihood by hand, and check the fit with standardized residuals

The parameters $\boldsymbol{\theta} = (g, \omega, \alpha, \beta)^\top$ are fitted by maximum likelihood (LM20). Given a start value σ_1^2, the recursion (26.3) turns each candidate $\boldsymbol{\theta}$ into a path $\sigma_1^2, \dots, \sigma_T^2$. For normal ξ_t the log-likelihood is LM20's normal one, with a variance that changes each day:

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2} \sum_{t=1}^{T} \left[\ln 2\pi + \ln \sigma_t^2 + \frac{(r_t - g)^2}{\sigma_t^2} \right]. \tag{26.5}$$

The arch package starts from its backcast, a mean of the first 75 squared residuals about the sample mean with weights proportional to 0.94^j, and sets $\sigma_1^2 = \omega + (\alpha + \beta) \times$ backcast. It maximizes (26.5) numerically. With Student t innovations the density becomes (15.9)'s, scaled by (15.11), and ν joins $\boldsymbol{\theta}$.

Classical standard errors invert $-\ell''$ as in LM20's (20.6). Robust ones use a sandwich that does not assume the density is right (arch's `cov_type` options `"classic"` and `"robust"`, its default). Their validity for large T under the dependence of (26.3) is stated without proof; LM20 gives the iid theory.

EXAMPLE 5

GARCH on TR23

Fit (26.3) to TR23's 5,849 log returns, in percent, with normal and with t innovations, and give the half-life with an interval.

SOLUTION

Exhibit 3 lists both fits. For the normal fit, the NumPy likelihood (26.5), started from arch's backcast of 2.78 %², equals arch's $\ell = -8{,}647.25$ to 10^{-6}. Its $\alpha + \beta = 0.98443$ gives $\sigma^2 = 1.597$ %², 19.83% a year (robust delta-method standard error 2.3 points) against the sample's 18.99% (LM19).

The half-life is 44.2 sessions. The delta method (20.8) with robust errors gives it a standard error of 16.1, so a 95% interval runs from 13 to 76 sessions. Mapping instead the robust interval for $\alpha + \beta$, $0.98443 \pm 1.96 \times 0.0056$ (0.97342 to 0.99544 unrounded), gives 26 to 152 sessions: near 1 the half-life is poorly determined.

With t innovations, $\hat{\nu} = 5.54$, ℓ rises by 183.35, and the half-life is 76.5 sessions; mapped from $\alpha + \beta$ (0.98343 to 0.99853), its interval runs from 41 to 471 sessions.

Exhibit 3: GARCH(1,1) fits to TR23

	Normal	Classical SE	Robust SE	Student t	Classical SE	Robust SE
g (% a day)	0.0809	0.0123	0.0126	0.0965	0.0114	0.0114
ω (%²)	0.0249	0.0040	0.0068	0.0149	0.0036	0.0043
α	0.0913	0.0076	0.0137	0.0663	0.0081	0.0104
β	0.8932	0.0086	0.0151	0.9247	0.0090	0.0117
ν	—			5.54	0.41	0.41
$\alpha + \beta$	0.98443			0.99098		
σ^2 (%²), a year	1.597, 19.83%			1.651, 20.16%		
Half-life (sessions)	44.2			76.5		
ℓ	−8,647.25			−8,463.90		

Note: TR23, 5,849 log returns in percent, in-sample; arch 8.0.0, constant mean, started from arch's backcast. Derived rows use unrounded estimates; $\alpha + \beta$ has five decimals so that each half-life follows from it by (26.4). Intervals for the derived rows: Example 5; the t fit's 20.16% has a standard error of 2.9 points. Source: as Exhibit 1.

EXAMPLE 6

Standardized residuals

Check the normal fit with its standardized residuals $\hat{\xi}_t = (r_t - \hat{g})/\hat{\sigma}_t$.

SOLUTION

A right model leaves $\hat{\xi}_t$ close to iid normal. The $\hat{\xi}_t$ have excess kurtosis 2.57 against r_t's 5.428 (LM17), and skewness −0.59 against r_t's −0.47 (computed here), both by (19.3).

The Ljung–Box statistic (25.5) of $|\hat{\xi}_t|$ on ten lags is 27.6 (p-value 0.002), against 1,820 for $|r_t|$. The ARCH test with $K = 5$ on $\hat{\xi}_t$ gives 14.3 (p-value 0.014). Both p-values are approximate, since the tests take residuals of an estimated model as data. GARCH removes most of the dependence between sizes, not all, and leaves fat tails.

LM5's lab asked, for 0050, whether changing volatility explains the tail beyond −4.107 standard deviations. On TR23, at or below −4.107 lie 17 days standardized by s and 9 standardized by GARCH, against 0.117 under the normal model. Changing volatility explains part of the tail, not all.

DEEP DIVE **Deep Dive 2: the kurtosis that GARCH implies** optional · click to expand

Let (26.3)'s conditions hold, with ε_t stationary and $\mathbb{E}\varepsilon_t^4 < \infty$. Square (26.3) and take expectations, using $\mathbb{E}\varepsilon_{t-1}^4 = \mathbb{E}\xi_t^4\,\mathbb{E}\sigma_{t-1}^4$ and $\mathbb{E}[\varepsilon_{t-1}^2\sigma_{t-1}^2] = \mathbb{E}\sigma_{t-1}^4$ (condition 1), $\mathbb{E}\sigma_t^4 = \mathbb{E}\sigma_{t-1}^4$ and $\omega = \sigma^2(1-\alpha-\beta)$:

$$\begin{aligned}\mathbb{E}\sigma_t^4\left(1-(\alpha+\beta)^2-(\mathbb{E}\xi_t^4-1)\alpha^2\right)\\ = \sigma^4\left(1-(\alpha+\beta)^2\right).\end{aligned}$$

Since $\mathbb{E}\varepsilon_t^4 = \mathbb{E}\xi_t^4\,\mathbb{E}\sigma_t^4$, when the bracket on the left is positive,

$$\operatorname{Kurt}(\varepsilon_t) = \frac{\mathbb{E}\xi_t^4\left(1-(\alpha+\beta)^2\right)}{1-(\alpha+\beta)^2-(\mathbb{E}\xi_t^4-1)\alpha^2}.$$

For normal ξ_t, $\mathbb{E}\xi_t^4 = 3$ by (14.6). At $\hat{\alpha} = 0.091254$ and $\hat{\alpha}+\hat{\beta} = 0.984432$ the denominator is 0.014239, so the kurtosis is 6.51, an excess of 3.51. For the t fit, $\mathbb{E}\xi_t^4 = 3 + 6/(\nu-4) = 6.89$ by (15.10) at $\hat{\nu} = 5.544$, and $(\alpha+\beta)^2 + (\mathbb{E}\xi_t^4-1)\alpha^2 = 1.008 > 1$: that model has no finite fourth moment.

EXAMPLE 7

What GARCH explains

How much of the clustering that LM13, LM15 and LM16 found does the normal fit reproduce?

SOLUTION

After the 108 falls of 3% or more, 13 next days fell too (LM13; LM21 tests it). GARCH, with the residuals' own distribution for ξ_t, expects 7.8, against 2.0 for independent days. After the 67 rises above 3% it expects 5.4, where 3 fell. The model raises the risk after any large move; the data raise it more after falls. GARCH(1,1) cannot express that asymmetry, which Hansen and Lunde (2005) call a "leverage effect"; the lab fits it.

After the 489 days with $|R_t| \geq 2\%$ the next day's size averaged 1.27%, against 0.80% after other days, LM16's pattern on TR23. GARCH predicts 1.35% and 0.79%.

Experiment 1 draws 1,000 paths of 5,849 days from the normal fit, with normal ξ_t and σ_t^2 started at the unconditional variance (`default_rng([1, 20261011])`). On TR23's 286 months, the variance-to-mean ratio of monthly counts of $|R| \geq 3\%$ has median 3.15 (batch standard error 0.035, LM18), with 5% to 95% of paths from 2.04 to 5.14. TR23 has 3.02 (LM15), and resampled days about 1 (LM18's 0.598 over LM15's mean, 0.612).

Experiment 1's paths have a median excess kurtosis of only 1.63, and 5% reach TR23's 5.428 (standard errors 0.045 and 0.7 points). Experiment 2 (`default_rng([2, 20261011])`) repeats it from the t fit: median 7.0 (standard error 0.11).

PITFALL **A robust standard error does not repair a wrong model**

The sandwich keeps the standard errors honest when ξ_t is not normal, but the estimates still belong to the fitted model. If the variance reacts more to falls than to rises, both kinds of error describe the wrong parameters.

導讀 **用最大概度估計配適 GARCH**

給定起始值之後，GARCH 的遞迴（recursion）把每一組參數（parameter）變成一條條件變異數（conditional variance）的路徑，概度函數（likelihood function）就是每天一個常態密度（normal density）的乘積，只是每天的變異數不同。最大概度估計（maximum likelihood estimation）找出讓這個乘積最大的參數。標準誤（standard error）有兩種：古典標準誤（classical standard error）假設模型的分布（distribution）正確；穩健標準誤（robust standard error）用夾層估計量（sandwich estimator），不需要這個假設。檢查配適（fit）時看標準化殘差（standardized residual）：加權股價報酬指數的標準化殘差還有超額峰態（excess kurtosis）2.57，GARCH 解釋了部分厚尾（heavy tail），但不是全部。

KNOWLEDGE CHECK 4

Why is the robust standard error of $\hat{\alpha}$ 1.80 times the classical one?

ANSWER

The classical error assumes that the normal density is right. The sandwich does not, and the standardized residuals of Example 6 have excess kurtosis 2.57, not 0.

6 REALIZED VARIANCE AND THE SIGNATURE PLOT

- [ ] compute realized variance from intraday data, read a volatility signature plot, and state the hypotheses (microstructure noise, stale prices) that could explain how it moves with the sampling interval

A day's variance is never observed, but intraday prices measure it. Take a grid of interval h from 09:30:00 to 13:30:00, with x_i the log index at point i. The **realized variance** of day t is LM3's RV on these intraday log returns. A return over Jh sums J returns over h, so for returns with mean zero and autocorrelations ρ_k that hold all day, (19.7) gives, ignoring the day's edges,

$$\begin{aligned} \mathrm{RV}_t(h) &= \sum_{i=1}^{n} (x_i - x_{i-1})^2, \quad n = \frac{14{,}400\ \mathrm{s}}{h}, \\ \frac{\mathbb{E}\,\mathrm{RV}(Jh)}{\mathbb{E}\,\mathrm{RV}(h)} &= 1 + 2\sum_{k=1}^{J-1} \left(1 - \frac{k}{J}\right)\rho_k. \end{aligned} \tag{26.6}$$

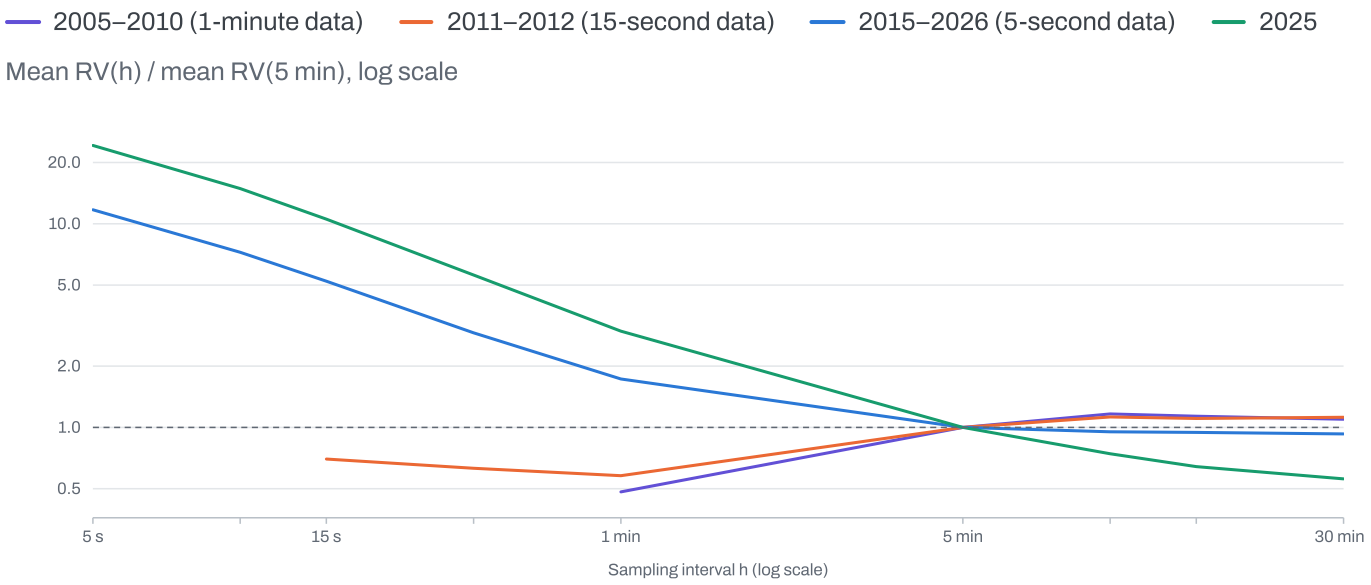
For $J = 2$ the ratio is $1 + \rho_1$.

The grid does not start at 09:00:00 because FinMind prints the previous close there, and the index needs about a minute to absorb the opening gap. It starts at 09:30:00 because the 240 minutes from there to 13:30:00 divide evenly into every interval used here, 5 seconds to 30 minutes.

The **opening return** $r_{\mathrm{o},t}$ runs from the previous close to 09:30:00, and the daily target $V_t := r_{\mathrm{o},t}^2 + \mathrm{RV}_t(15\ \mathrm{min})$, in %², splits the day into 17 squared pieces. A day has 2,880 five-second returns, or 16 fifteen-minute ones. Dividends change TR23's squared returns by less than 0.01% in total on these days.

Without noise, a finer grid only helps: if day t's n returns were iid normal with variance σ_t^2/n each, $\mathrm{RV}_t(h)$ would have mean σ_t^2 and variance $2\sigma_t^4/n$ by (14.6). Exhibit 4 plots the **volatility signature**, a period's mean $\mathrm{RV}(h)$ over its mean $\mathrm{RV}(5\ \mathrm{min})$. On the 5-second days of 2015–2026, $\mathrm{RV}(5\ \mathrm{s})$ averaged 11.7 times $\mathrm{RV}(5\ \mathrm{min})$, and in 2025, 24.3 times. In 2005–2010 $\mathrm{RV}(1\ \mathrm{min})$ averaged only 0.483 of it.

Exhibit 4: The volatility signature of the TAIEX



Note: the TAIEX price index, 09:30–13:30; log scales. Data every minute in 2005–2010 (1,493 days), every 15 seconds from 17 January 2011 to 2012 (487 days) and every 5 seconds in 2015–2026 (2,867 days, 2025 included). The 2013–2014 sessions are in Exhibit 5. Descriptive. Source: FinMind `TaiwanVariousIndicators5Seconds` (TWSE's TAIEX), retrieved 19 October 2026. Computations: `data/lm26_numbers.py`.

EXAMPLE 8

The variance ratio read backwards

By (26.6), $\mathrm{RV}(2h)/\mathrm{RV}(h) - 1$ estimates ρ_1 of returns over h. Compare it with the pooled $\hat{\rho}_1$ of the raw download, within-day pairs only.

SOLUTION

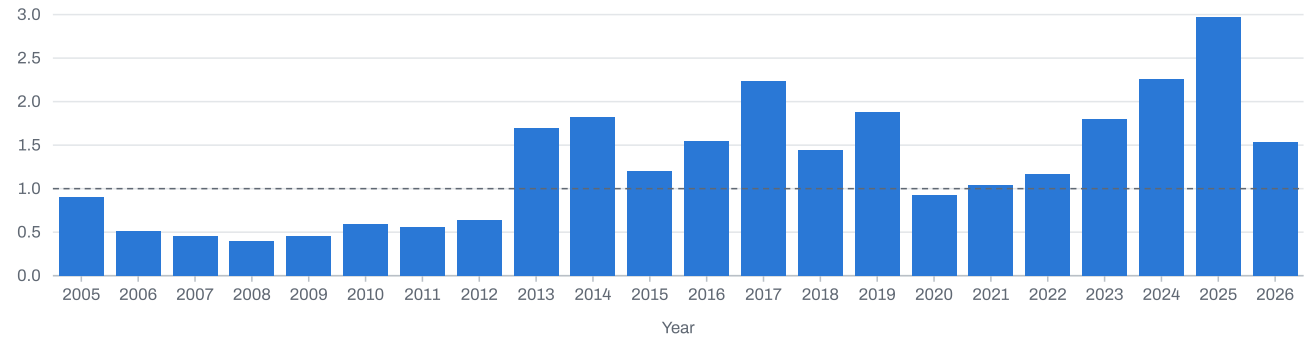
On the 5-second days, $\mathrm{RV}(10\text{ s})/\mathrm{RV}(5\text{ s}) - 1 = -0.381$ from the per-day file (a ratio of the period's means), against a pooled $\hat{\rho}_1 = -0.380$ of 5-second returns. On 2011–2012's 15-second days the two are −0.099 and −0.0992, and at 5 minutes over all days 0.056 both ways. Five-second returns reverse: a rise tends to be followed by a fall.

In 2025 the pooled $\hat{\rho}_1$ of 5-second returns was −0.386 (raw download). Read through the noise model (26.7) below, a hypothesis, noise then supplies 77% of a 5-second return's variance. Each day's own $\hat{\rho}_1$ ran from −0.52 to −0.19, the middle half from −0.41 to −0.33. The ratio $\mathrm{RV}(5\text{ s})/\mathrm{RV}(5\text{ min})$ was 25.2 on the median day and up to 62.6.

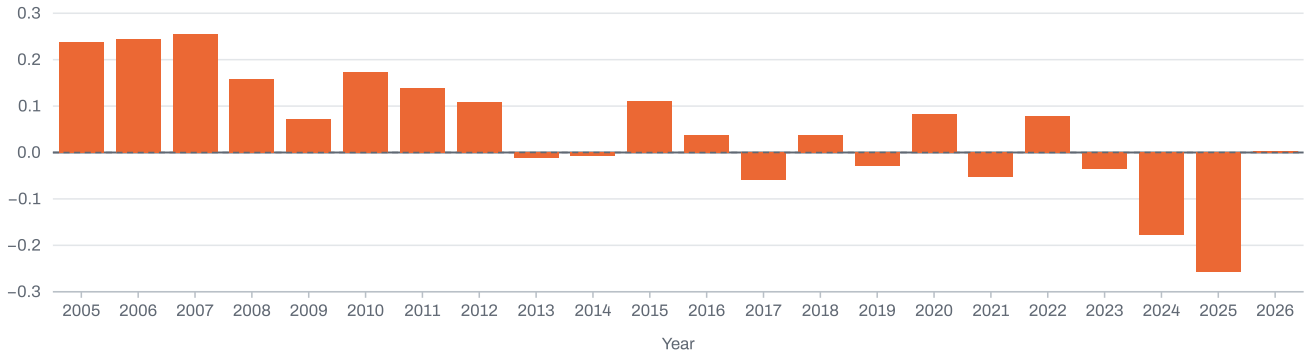
Exhibit 5 shows the change by year. By (26.6) with $J = 5$, a top bar below 1 says that 1-minute returns within five minutes correlate positively on balance. The bars run from 0.397 in 2008 to 2.968 in 2025, below 1 in 2005–2012 and 2020 only. The bottom bars approximate $\hat{\rho}_1$ of 5-minute returns ($J = 2$): from 0.256 in 2007 to −0.257 in 2025, below −0.1 only in 2024 (−0.177) and 2025.

Exhibit 5: Ratios of realized variances at adjacent intervals, by year

RV(1 min) ÷ RV(5 min)



RV(10 min) ÷ RV(5 min) − 1



Note: the TAIEX price index, 09:30–13:30, ratios of a year's mean realized variances. Neither is itself an autocorrelation; over all days the bottom ratio and the pooled $\hat{\rho}_1$ of 5-minute returns are both 0.056. 2026 runs to 8 October. Descriptive. Source: as Exhibit 4.

One model of the reversal separates the observed price from an efficient one.

RESULT **A noise model (a hypothesis)**

Let $x_i = x_i^{\mathrm{e}} + u_i$, where the efficient returns $x_i^{\mathrm{e}} - x_{i-1}^{\mathrm{e}}$ are iid with variance σ_{e}^2, the noise u_i is iid with variance σ_u^2, and the two are independent. Then the log index is LM25's persistent part plus noise, (25.9), with a random walk as the persistent part, and its returns form an MA(1), (25.3). Their lag-1 autocorrelation and mean realized variance are

$$\begin{aligned} \rho_1 &= -\frac{\sigma_u^2}{\sigma_{\mathrm{e}}^2 + 2\sigma_u^2} \in \left[-\tfrac{1}{2}, 0\right], \\ \mathbb{E}\,\mathrm{RV}(h) &= n\sigma_{\mathrm{e}}^2 + 2n\sigma_u^2. \end{aligned} \tag{26.7}$$

Here $n\sigma_{\mathrm{e}}^2$ is the window's efficient variance, the same for every h, while the noise term grows with n.

DEEP DIVE **Deep Dive 3: noise and the variance ratio** optional · click to expand

Each observed return is $(x_i^{\mathrm{e}} - x_{i-1}^{\mathrm{e}}) + u_i - u_{i-1}$, with variance $\sigma_{\mathrm{e}}^2 + 2\sigma_u^2$. Adjacent returns share u_i with opposite signs, so their covariance is $-\sigma_u^2$; returns further apart share nothing. That gives ρ_1 of (26.7), the MA(1) shape of (25.3), and $\mathbb{E}\,\mathrm{RV}(h)$ as a sum of n variances. The noise share of a return's variance is $2\sigma_u^2/(\sigma_{\mathrm{e}}^2 + 2\sigma_u^2) = 2|\rho_1|$.

If (26.7) held, the mean RV would rise by the same $2\sigma_u^2$ for every extra return a day. On the 5-second days, between neighbouring intervals from 5 seconds to 5 minutes, the rise per extra return runs from 3.1×10^{-3} to 4.9×10^{-3} of $\mathrm{RV}(5\ \mathrm{min})$. Each has a Newey–West standard error across days of about a tenth of itself. Day by day the two extremes move together (correlation 0.93), and the Newey–West t-stat of their per-day difference is 10.6.

The Newey–West errors need a finite fourth moment of returns (LM19), so they are nominal. Yet the largest rise exceeds the smallest on 88% of the days.

So (26.7), with iid efficient returns and iid noise, does not fit. In the 30 Dow Jones stocks, Hansen and Lunde (2006) find noise that is serially dependent and correlated with the efficient returns.

EXAMPLE 9

The index against its future

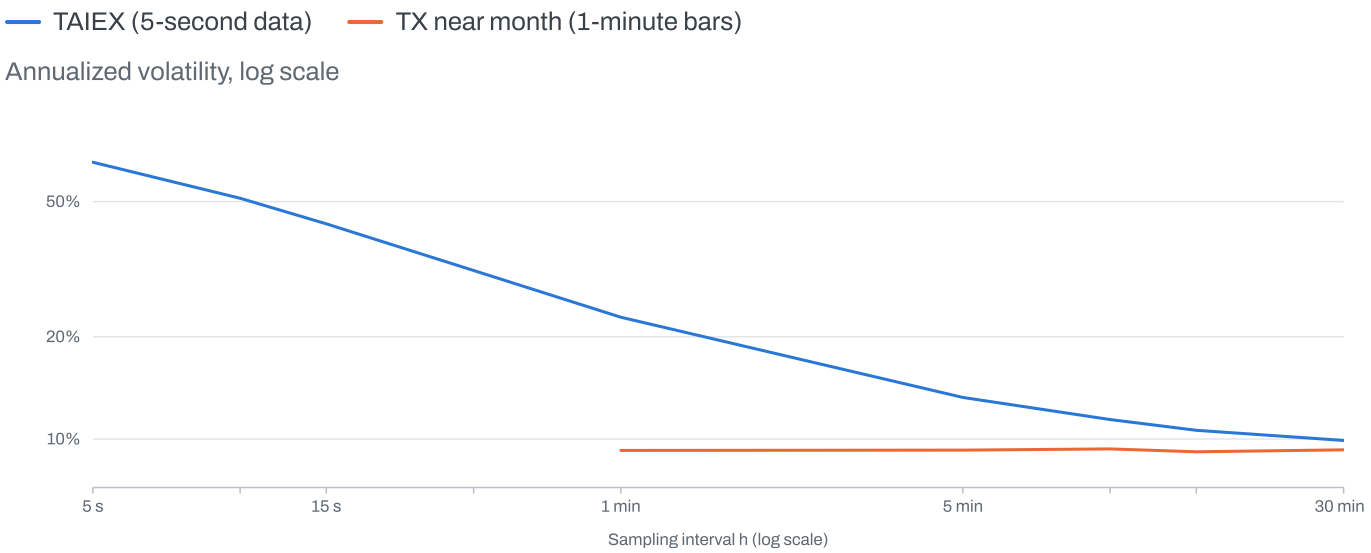
Compare the TAIEX with the near-month TX future in 2025, on the same window and intervals.

SOLUTION

In Exhibit 6, the TAIEX's 1-minute returns have $\hat{\rho}_1 = -0.421$ (raw download) and a window volatility of 22.8% a year. TX's 1-minute bars give −0.011 and 9.26%, and TX's ratios to 5 minutes are 1.00, 1.02, 0.98 and 1.00 at 1, 10, 15 and 30 minutes: a flat signature.

Whatever moves the index at high frequency is not in the futures price. Candidate causes, such as slow stocks' prices updating late and heavy stocks bouncing between bid and ask, are hypotheses: no source in this book states either for the TAIEX.

Exhibit 6: TAIEX and TX volatility by sampling interval, 2025



Note: 243 sessions of 2025, 09:30–13:30, $\sqrt{A \times \text{mean RV}(h)}$; log scales. TX: the near-month contract, from 1-minute bars labelled by their first minute (an inference from the labels), so 09:30:00 is the 09:29 bar's close. The limit-locked 7 and 10 April leave TX's ratios unchanged to four decimals. The index's variance is 6.08, 2.04, 1.34 and 1.14 times TX's at 1, 5, 15 and 30 minutes. Source: as Exhibit 4, and FinMind `TaiwanFuturesKBar` (TX) and `TaiwanFuturesDaily` (TX, the contract held), retrieved 19 October 2026.

In the raw download the index freezes on every day from the first value after 13:25:00 to the last before 13:30:00, the closing call auction. The move to the close lands in one 5-minute return, 7.3% of $\text{RV}(5\text{ min})$ on the 5-second days against $1/48 = 2.1\%$ on average. And the opening return supplies 66% of the mean V_t over the 5,351 days.

The target uses 15 minutes because the signature flattens near there in most years; Section 7's ranking is the same at 5 and 30 minutes. The book does not cover jumps, or estimators built to withstand noise such as the two-scale realized variance of Zhang, Mykland and Aït-Sahalia (2005, §1.2.5) and realized kernels.

> **PITFALL** **The finest interval is not the best**
>
> Read through (26.7), noise supplies most of a 5-second return's variance in 2025. Even at 5 minutes the index carried twice TX's variance that year. Choose the interval from the signature, not from the finest data available.

導讀 **最細的資料不一定最好**

實現變異數（realized variance）是把一天之內的對數報酬（log return）平方加總，開根號就是實現波動率（realized volatility）。如果價格沒有雜訊（noise），抽樣越細越準；但用加權指數每 5 秒一筆的高頻資料（high-frequency data），2025 年算出的實現變異數是 5 分鐘的 24 倍，同期台指期卻幾乎不隨抽樣區間（sampling interval）變動。相鄰的 5 秒報酬強烈負自我相關（autocorrelation），一種解釋是市場微結構雜訊（market microstructure noise）：觀察到的價格等於有效價格（efficient price）加上雜訊。這只是假說，股價更新延遲、在買賣價（bid and ask）之間跳動等成因，本書沒有文獻可以確認。本模組的目標值（target）取 15 分鐘的實現變異數，再加上開盤報酬（opening return）的平方。

KNOWLEDGE CHECK 5

In 2011–2012, $\mathrm{RV}(30\text{ s})/\mathrm{RV}(15\text{ s}) = 0.9013$. What pooled $\hat{\rho}_1$ of 15-second returns does (26.6) imply?

ANSWER

With $J = 2$, $1 + \rho_1 \approx 0.9013$, so $\hat{\rho}_1 \approx -0.099$. The raw download gives −0.0992.

7 JUDGING FORECASTS OUT OF SAMPLE

- ☐ evaluate variance forecasts out of sample with MSE and QLIKE against a noisy realized-variance target, and with Mincer–Zarnowitz regressions

Six forecasts are fixed in advance and made each evening for the next session. GARCH with normal and with t innovations is refitted at each month's first session on the previous 1,231 returns, five years by LM17's rule (226 refits). The others are EWMA, the 60-day mean of squared returns, yesterday's V, and an EWMA of V, both EWMAs with $\lambda = 0.94$. Forecasts run from 2 January 2008 to 8 October 2026: 4,609 days.

DEFINITION **Losses for a variance forecast**

For a target $V_t > 0$ and a forecast $\sigma^2_{t|t-1} > 0$, the losses of day t are

$$\begin{aligned} \text{squared error} &= \left(V_t - \sigma^2_{t|t-1}\right)^2, \\ \text{QLIKE} &= \frac{V_t}{\sigma^2_{t|t-1}} - \ln \frac{V_t}{\sigma^2_{t|t-1}} - 1 \;\geq 0. \end{aligned} \tag{26.8}$$

A forecast's MSE is the mean of its squared errors over the test days, in $\%^4$, and its QLIKE score the mean of its daily QLIKE. QLIKE has no unit and is zero for a perfect forecast.

The target V_t is only a noisy measurement of the variance. MSE and QLIKE still rank forecasts as the true variance would if $\mathbb{E}[V_t \mid \text{day } t-1] = \sigma^2_t$, a conditionally unbiased target. That result is stated without proof (Patton 2011; his 2006 working paper, Section 3). QLIKE depends only on the ratio of target to forecast, so a calm day counts as much as a crash.

DEFINITION **Mincer–Zarnowitz regression**

Regress the target on a constant and the forecast by OLS, (23.1):

$$\mathbb{E}\left[V_t \mid \text{day } t-1\right] = a + b\,\sigma^2_{t|t-1}. \tag{26.9}$$

A conditionally unbiased forecast has $a = 0$ and $b = 1$. The target's errors grow with the forecast, so the standard errors are White's HC0, (23.6).

EXAMPLE 10

Six forecasts, 2008–2026

Rank the six forecasts against V_t (Exhibit 7), test the differences and check the best one for bias.

SOLUTION

GARCH normal has the lowest QLIKE, 0.5032, and the lowest MSE, 10.37 $\%^4$. To test a difference, take each day's loss minus GARCH normal's. Its mean over its Newey–West standard error, (23.7) with 9 lags for 4,609 days, is a t-stat with a normal reference. EWMA is worse by 0.0121 in QLIKE (t-stat 1.83, p-value 0.067) and by 0.428 in MSE (t-stat 2.98). GARCH t differs by 1.40 and 0.62 standard errors; yesterday's V is worst (t-stat 11.96 in QLIKE).

For GARCH normal, (26.9) gives $a = -0.015$ (HC0 standard error 0.135) and $b = 0.980$ (0.114), so the t-stat for $b = 1$ is −0.18; R-squared is 0.172, and 0.119 against r^2_t. Andersen and Bollerslev (1998) found 0.047 and 0.026 against squared returns for two exchange rates, and about half against 5-minute realized volatility. Our target lies between, because over the 4,609 days 68% of it is, on average, one squared opening return.

Exhibit 7: Six variance forecasts against the target V_t, 2008–2026

Forecast	Mean (%²)	MSE (%⁴)	QLIKE	b (HC0 SE)	R-squared	t-stat, QLIKE
Rolling 60-day	1.466	11.48	0.5774	0.763 (0.059)	0.093	4.82
EWMA 0.94	1.464	10.80	0.5153	0.850 (0.083)	0.143	1.83
GARCH normal	1.411	10.37	0.5032	0.980 (0.114)	0.172	—
GARCH t	1.394	10.41	0.5077	1.010 (0.118)	0.168	1.40
Yesterday's V	1.368	18.18	1.1778	0.274 (0.054)	0.075	11.96
EWMA of V	1.370	10.78	0.5209	0.864 (0.074)	0.143	1.46

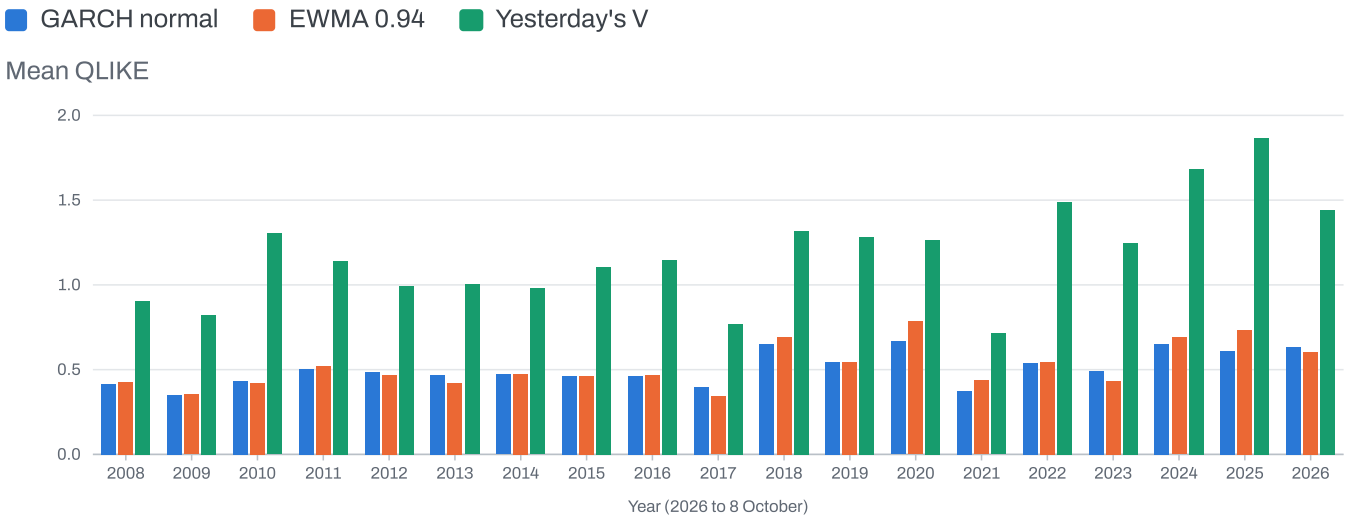
Note: 4,609 out-of-sample days, 2 January 2008 to 8 October 2026; target V_t, mean 1.368 %². Slope b of (26.9); t-stat of the mean QLIKE difference from GARCH normal (Newey–West). The QLIKE order is the same with $V_t(5\ \text{min})$ or $V_t(30\ \text{min})$ as the target; with r_t^2 only the two EWMAs swap places. Source: as Exhibit 1.

The target itself falls short of the close-to-close day: it averages 0.933 of r_t^2 over the 5,351 days (Newey–West t-stat −3.26), 0.918 in 2025 and 1.011 in 2008. So the QLIKE order is also checked against r_t^2, unbiased up to the tiny squared mean; there only the two EWMAs swap places.

The MSE t-stats, the HC0 errors of b and the t-stat −3.26 need a finite fourth moment of returns, which LM15's fit lacks (LM19); they are nominal.

Exhibit 8 splits QLIKE by year. GARCH beats EWMA in 12 of 19 years; in 2025 it scored 0.611 against 0.733, and in 2020 0.666 against 0.785. With windows of 1,000 and 2,461 returns instead of 1,231, GARCH normal again has the lowest QLIKE, 0.5029 and 0.5332, against EWMA's 0.5126 and 0.5421. Those forecasts run from 1 February 2007 and from 3 December 2012.

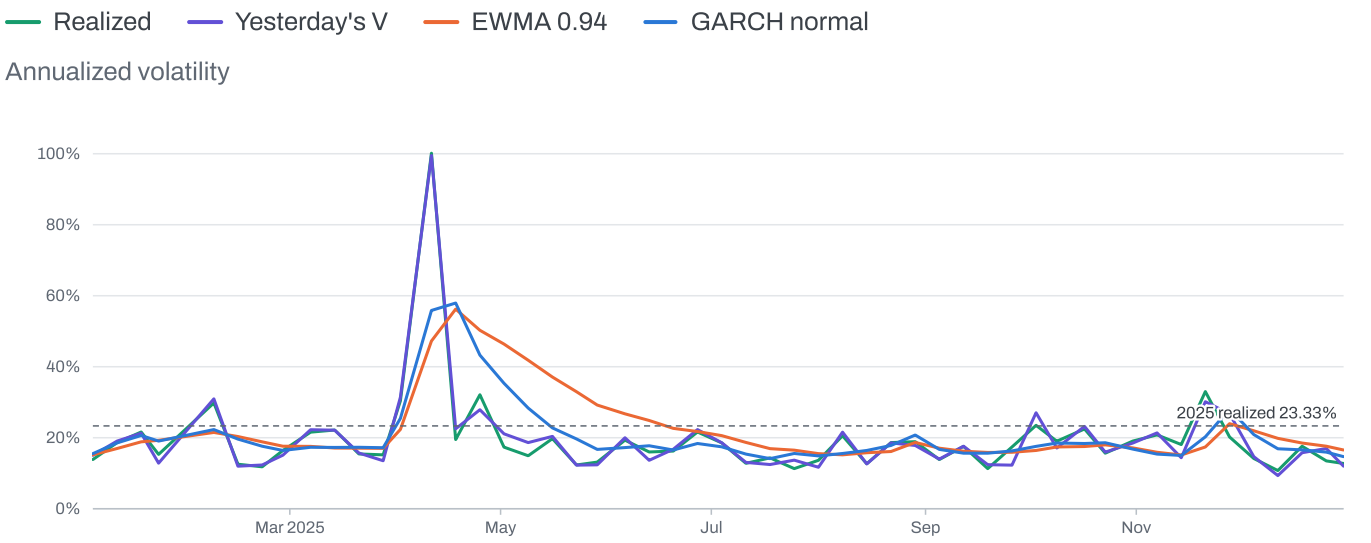
Exhibit 8: QLIKE by year



Note: mean QLIKE (26.8) against V_t in each calendar year, out of sample; 2026 runs to 8 October (185 days). Source: as Exhibit 1.

Exhibit 9 follows 2025, whose realized volatility was 23.33%, week by week. In the week of 7 April it was 100.16% a year; GARCH forecast 55.86% on average, EWMA 47.22% and yesterday's V 99.52%. Over the 20 sessions from 7 April it was 54.3%, against 19.8% in the 20 before; GARCH forecast 48.9%, EWMA 50.1%. Neither forecast wins every stretch.

Exhibit 9: Realized and forecast volatility by week, 2025



Note: the 243 sessions of 2025 in 52 ISO weeks, the last holding 29–31 December. Each point, at the week's last session, is $\sqrt{A \times \text{the week's mean}}$ of V_t or of a forecast. Out of sample. Source: as Exhibit 1.

Forecasting a whole covariance matrix is LM30's and LM31's subject.

PITFALL **A noisy, unbiased target lowers R-squared, not the ranking**

GARCH's R-squared of 0.119 against r_t^2 does not make it a poor forecast: a single squared return is a noisy target, and its noise caps R-squared. Given the past, r_t^2 is unbiased up to the tiny squared mean, so a robust loss still ranks forecasts as the true variance would.

導讀 **用有雜訊的目標值評估預測**

真正的每日變異數（variance）看不到，只能拿實現變異數（realized variance）這類有雜訊（noise）的量來比，所以損失函數（loss function）的選擇很重要。Patton 證明，只要目標值（target）條件不偏（conditionally unbiased），均方誤差（mean squared error，MSE）與 QLIKE 的排序會和用真實變異數排序相同。QLIKE 只看目標值與預測值（forecast）的比率，對預測偏低的懲罰比偏高重。在 4,609 天的樣本外預測（out-of-sample forecasting）中，GARCH 的 QLIKE 最低，但只比 EWMA 好 1.83 個標準誤（standard error）。條件不偏的目標值，其雜訊會壓低判定係數（R-squared），卻不會改變排序。

KNOWLEDGE CHECK 6

The target is $1\ \%^2$. What QLIKE does a forecast of half the target score, and one of double the target? What squared error does each score?

ANSWER

Half: $2 - \ln 2 - 1 = 0.3069$. Double: $0.5 + \ln 2 - 1 = 0.1931$. The squared errors are 0.25 and 1. For errors of the same ratio, QLIKE punishes the forecast that is too low more, and the squared error the one that is too high. The squared error itself is symmetric in $V_t - \sigma^2_{t|t-1}$.

8 VOLATILITY MODELS IN PYTHON

- [ ] run these tools in Python with arch and statsmodels on NumPy arrays, matching each package result to a calculation by hand

PYTHON

```python
import numpy as np
import polars as pl
from arch import arch_model
from statsmodels.stats.diagnostic import acorr_ljungbox, het_arch

tr = pl.read_csv("data/taiex_tr.csv")["price"].to_numpy()
y = 100 * np.log1p(tr[1:] / tr[:-1] - 1)      # log returns in percent
res = arch_model(y, mean="Constant", vol="GARCH", p=1, q=1).fit(
    disp="off", cov_type="classic")
g, omega, alpha, beta = res.params.to_numpy()  # a pandas Series in arch
fc = res.forecast(horizon=250, reindex=False).variance.to_numpy()[0]
xi = (y - g) / res.conditional_volatility      # an array for array input
arch5 = het_arch(y - y.mean(), nlags=5, result_object=True)
print(arch5.lm)                                # 696.3
print(acorr_ljungbox(np.abs(xi), lags=[10]).to_numpy())   # 27.6
rv = pl.read_csv("data/vol4_rv.csv", comment_prefix="#")
V = 1e4 * (rv["o2"] + rv["rv_15m"]).to_numpy()  # the target, %^2
```

`het_arch` squares the residuals it is given, so pass $r_t - \bar{r}$; without `result_object=True` it returns a tuple and warns. `acorr_ljungbox` returns a DataFrame. `comment_prefix="#"` skips the derived file's first line, a comment.

IN PYTHON **Companion lab**

The Polars-first notebook `LM26_lab.ipynb` rebuilds the ARCH tests, the EWMA and GARCH forecasts, the fits by hand and with arch, the simulations, the signature plots and the out-of-sample comparison. It solves the computational practice problems and ends with an asymmetric GARCH.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline. The realized variances come with the repository (`data/derived/vol4_rv.csv`). Rebuilding them needs the 5-second TAIEX download, `uv run python data/fetch_intraday.py --series taiex5s` (5,351 requests, about nine hours on a free registered FinMind account at 600 calls an hour, re-run after each hourly stop).

KNOWLEDGE CHECK 7

Why does the book give arch $100r_t$ rather than r_t, and what does `.to_numpy()` do to `res.params`?

ANSWER

The arch package warns when the residual variance lies outside [0.1, 10,000): TR23's daily variance is about 0.000147 in decimals and 1.47 in $\%^2$. `.to_numpy()` turns arch's pandas Series into a NumPy array in arch's order (g, ω, α, β).

SUMMARY

- Returns are nearly uncorrelated, their sizes are not; the ARCH test (26.1) detects that but names no model.
- EWMA (26.2) weights squared returns by $(1-\lambda)\lambda^j$; RiskMetrics uses $\lambda = 0.94$.
- GARCH(1,1) (26.3) pulls forecasts toward σ^2 with half-life $\ln\frac{1}{2} / \ln(\alpha+\beta)$, (26.4): 44.2 sessions on TR23.
- The likelihood (26.5) follows from the recursion; for ω, α and β of the normal fit, robust errors are 1.7 to 1.8 times the classical ones.
- Standardized residuals keep fat tails; GARCH(1,1) misses the asymmetry after falls.
- Realized variance (26.6) depends on the interval; under the noise model (26.7) fine returns reverse.
- MSE and QLIKE (26.8) rank forecasts as the true variance would if the target is conditionally unbiased; GARCH leads, but not every year.

PRACTICE PROBLEMS

Problems on data use TR23's 5,849 log returns in percent, $A = 246.12$, moments with divisor T, and Section 7's design and target.

1. An EWMA has $\lambda = 0.97$. Its effective window is *closest* to:

   A. 22.76 days

   B. 33.3 days

   C. 32.3 days

2. A GARCH(1,1) has $\omega = 0.025\ \%^2$, $\alpha = 0.09$ and $\beta = 0.89$ (constructed), on daily returns in percent. With $A = 246.12$, its unconditional annualized volatility is *closest* to:

   A. 17.5%

   B. 7.5%

   C. 1.12%

3. A GARCH(1,1) has $\alpha + \beta = 0.98$. The half-life of a shock to its variance forecast is *closest* to:

   A. 25 sessions

   B. 50 sessions

   C. 34.3 sessions

4. Judged against r_t^2 instead of V_t, GARCH normal's Mincer–Zarnowitz R-squared falls from 0.172 to 0.119, while the two GARCH forecasts keep the two lowest QLIKE scores. The best explanation is:

   A. The target r_t^2 is conditionally unbiased but noisier: the ranking stays valid, and differences are harder to detect.

   B. The target r_t^2 is biased, so its ranking is invalid.

   C. A lower R-squared shows that GARCH forecasts worse than it seemed.

5. Five-second returns have a pooled $\hat{\rho}_1 = -0.40$ (constructed) and no other autocorrelation. The ratio $\mathrm{RV}(10\text{ s})/\mathrm{RV}(5\text{ s})$ is *closest* to:

   A. 1.40

   B. 0.60

   C. 0.20

6. Derive the forecast formula of (26.4) from (26.3). Apply it from the evening of 7 April 2025, with $\sigma^2_{t+1|t} = 12.3695\ \%^2$, $\alpha + \beta = 0.984432$ and $\sigma^2 = 1.5971\ \%^2$. Give $\sigma_{t+20|t}$ to three decimals, and the volatility of the average forecast variance over the next 20 days to one decimal, a year.

7. Show that EWMA's weights $(1-\lambda)\lambda^j$, $j \geq 0$, sum to 1, and that its effective window is $1/(1-\lambda)$. Show that EWMA is GARCH(1,1) with $\omega = 0$, $\alpha = 1-\lambda$ and $\beta = \lambda$, and that its forecast is the same for every horizon.

8. (a) From (19.7), derive $\mathbb{E}\,\mathrm{RV}(2h) = (1+\rho_1)\,\mathbb{E}\,\mathrm{RV}(h)$ for returns with mean zero and a common ρ_1 within the day, ignoring the day's edges. (b) Under (26.7), show that $\rho_1 \geq -\frac{1}{2}$ and that noise supplies the share $2|\rho_1|$ of a return's variance. (c) Apply (b) to 2025's pooled $\hat{\rho}_1 = -0.386$ of 5-second returns, to the nearest percent.

9. Derive the kurtosis of Deep Dive 2 for normal ξ_t, and evaluate it to two decimals at $\hat{\alpha} = 0.091254$ and $\hat{\alpha} + \hat{\beta} = 0.984432$. Explain why experiment 1's paths have a median excess kurtosis of only 1.63.

10. *(Python)* Compute the normal GARCH log-likelihood (26.5) in NumPy at arch's estimates, from arch's backcast (about the sample mean, 2.78 %²), and compare it with arch's. Report ℓ to two decimals and $\hat{\sigma}_t$ on 7 April 2025 to three.

11. *(Python)* Rebuild Section 7's EWMA (0.94) and GARCH-normal forecasts, with monthly refits on 1,231 returns. Compute their mean QLIKE against V_t to four decimals, and the Newey–West t-stat of the difference with 9 lags to two.

12. *(Python)* From `vol4_rv.csv`, rebuild the 2015–2026 line of Exhibit 4. Then estimate $\hat{\rho}_1$ of 5-second returns on the 5-second days by the ratio of the period's mean $\mathrm{RV}(10\text{ s})$ to its mean $\mathrm{RV}(5\text{ s})$, minus 1, to three decimals.

SOLUTIONS

1. **B** is correct: $1/(1-0.97) = 33.3$ days. A is the weights' half-life, $\ln\frac{1}{2}/\ln 0.97$. C is the mean lag $\lambda/(1-\lambda)$, one day shorter.

2. **A** is correct: $\sigma^2 = 0.025/(1-0.98) = 1.25$ %², and $\sqrt{246.12 \times 1.25} = 17.5\%$. B forgets α: $0.025/(1-0.89) = 0.2273$ %², 7.5% a year. C is the daily $\sqrt{1.25} = 1.12\%$, not annualized.

3. **C** is correct: $\ln\frac{1}{2}/\ln 0.98 = 34.3$ sessions by (26.4). A assumes a linear decay, $0.5/(1-0.98)$. B is $1/(1-0.98)$, the decay's effective window (Section 3), not its half-life.

4. **A** is correct: given the past, r_t^2 has mean σ_t^2 plus the tiny squared mean, so Patton's result keeps the ranking, while the noise lowers R-squared. B confuses noise with bias. C reads R-squared as accuracy, although the target's noise caps it.

5. **B** is correct: by (26.6) with $J = 2$, the ratio is $1 + \rho_1 = 0.60$. A reverses the sign. C counts the pair twice, $1 + 2\rho_1$.

6. Given day t, condition 1 and (16.5) give $\mathbb{E}[\varepsilon_{t+j}^2 \mid \text{day } t] = \sigma_{t+j|t}^2$ for $j \geq 1$. So (26.3) gives $\sigma_{t+j+1|t}^2 - \sigma^2 = (\alpha+\beta)(\sigma_{t+j|t}^2 - \sigma^2)$, and $m-1$ steps give (26.4). Then

$$\begin{aligned}\sigma_{t+20|t}^2 &= 1.5971 + 0.984432^{19} \\ &\quad \times (12.3695 - 1.5971) \\ &= 9.5925,\end{aligned}$$

so $\sigma_{t+20|t} = 3.097\%$. The average over $m = 1, \ldots, 20$ is

$$\begin{aligned}&1.5971 + 10.7724 \times \frac{1 - 0.984432^{20}}{20 \times 0.015568} \\ &= 10.9157\ \%^2,\end{aligned}$$

and $\sqrt{246.12 \times 10.9157} = 51.8\%$ a year.

7. The weights form a geometric series: $(1-\lambda)\sum_{j\geq 0}\lambda^j = (1-\lambda)/(1-\lambda) = 1$. Equal weights over $1/(1-\lambda)$ days also give today $1-\lambda$. With $\omega = 0$, $\alpha = 1-\lambda$ and $\beta = \lambda$, (26.3) is (26.2) for returns with mean zero. Then $\alpha + \beta = 1$, so the step of Problem 6 gives $\sigma_{t+j+1|t}^2 = \sigma_{t+j|t}^2$: every horizon has tomorrow's forecast.

8. (a) A return over $2h$ sums two adjacent returns over h, so by (19.7) its variance is $2\gamma_0(1+\rho_1)$. A day has $n/2$ such returns, so $\mathbb{E}\,\mathrm{RV}(2h) = n\gamma_0(1+\rho_1) = (1+\rho_1)\,\mathbb{E}\,\mathrm{RV}(h)$. (b) By Deep Dive 3, $\rho_1 = -\sigma_u^2/(\sigma_{\mathrm{e}}^2 + 2\sigma_u^2)$. Since $\sigma_{\mathrm{e}}^2 \geq 0$, the denominator is at least $2\sigma_u^2$, so $\rho_1 \geq -\frac{1}{2}$. The noise part of a return, $u_i - u_{i-1}$, has variance $2\sigma_u^2$, a share $2\sigma_u^2/(\sigma_{\mathrm{e}}^2 + 2\sigma_u^2) = 2|\rho_1|$. (c) Here $2 \times 0.386 = 0.772$: if (26.7) held, noise would supply 77% of a 5-second return's variance in 2025.

9. Deep Dive 2 gives the derivation. At the fit the denominator is $1 - 0.984432^2 - 2 \times 0.091254^2 = 0.014239$, and $\mathrm{Kurt} = 3 \times (1 - 0.984432^2)/0.014239 = 6.51$, an excess of 3.51. That is a population value. With so small a denominator, rare stretches of very high variance carry it, and 5,849 days usually miss them: the median path shows 1.63.

10. By hand $\ell = -8{,}647.25$, equal to arch's to 10^{-6}, and $\hat{\sigma}_t = 1.737\%$ on 7 April 2025 (Example 5; the lab).

11. The lab gives QLIKE 0.5153 for EWMA and 0.5032 for GARCH normal, and a Newey–West t-stat of 1.83 for the difference (Example 10).

12. The lab rebuilds the line of Exhibit 4, from 11.71 at 5 seconds to 0.930 at 30 minutes, and gives $\hat{\rho}_1 \approx -0.381$ (Example 8).

GLOSSARY

Term	中文	Meaning
Volatility clustering	波動群聚現象	Dependence between the sizes of returns: large moves follow large moves.
Conditional variance	條件變異數	σ_t^2, the variance of day t given the returns before it.
ARCH test	自我迴歸條件變異數非齊一檢定	$(T-K)\times$ R-squared of squared residuals on K lags, approximately χ^2_K for iid returns with a finite fourth moment, (26.1).
EWMA	指數加權移動平均模型	The forecast $\lambda\sigma^2_{t\|t-1}+(1-\lambda)r_t^2$, (26.2).
Effective window	有效視窗	$1/(1-\lambda)$, the number of equal weights that give today EWMA's weight.
GARCH(1,1)	廣義自我迴歸條件變異數非齊一模型	$\sigma_t^2=\omega+\alpha\varepsilon_{t-1}^2+\beta\sigma_{t-1}^2$ with (26.3)'s conditions.
Unconditional variance	無條件變異數	$\sigma^2=\omega/(1-\alpha-\beta)$, the level GARCH forecasts return to, (26.4).
Half-life	半衰期	$\ln\frac{1}{2}/\ln(\alpha+\beta)$ sessions, in which a shock's effect on the forecast halves.
Standardized residual	標準化殘差	$\hat\xi_t=(r_t-\hat g)/\hat\sigma_t$; close to iid if the model is right.
Realized variance	實現變異數	$\mathrm{RV}_t(h)$, the sum of squared intraday log returns at interval h, (26.6).
Realized volatility	實現波動率	The square root of a realized variance.
Volatility signature	—	Mean realized variance against the sampling interval, as a ratio to 5 minutes.
Market microstructure noise	市場微結構雜訊	The gap between the observed and an efficient price in (26.7), a hypothesis.

Term	中文	Meaning
Opening return	開盤報酬	$r_{\mathrm{o},t}$, the log change from the previous close to 09:30:00.
Loss function	損失函數	The squared error or QLIKE of a variance forecast against a target, (26.8).
Mincer–Zarnowitz regression	—	OLS of the target on a constant and the forecast, (26.9).
Out-of-sample forecast	樣本外預測	A forecast made only from data before the day it forecasts.

REFERENCES

- Andersen, T. G., and T. Bollerslev (1998). "Answering the Skeptics: Yes, Standard Volatility Models do Provide Accurate Forecasts." *International Economic Review* 39 (4): 885–905.
- Bollerslev, T. (1986). "Generalized Autoregressive Conditional Heteroskedasticity." *Journal of Econometrics* 31 (3): 307–327.
- Engle, R. F. (1982). "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation." *Econometrica* 50 (4): 987–1007.
- Hansen, P. R., and A. Lunde (2005). "A Forecast Comparison of Volatility Models: Does Anything Beat a GARCH(1,1)?" *Journal of Applied Econometrics* 20 (7): 873–889.
- Hansen, P. R., and A. Lunde (2006). "Realized Variance and Market Microstructure Noise." *Journal of Business & Economic Statistics* 24 (2): 127–161.
- J.P. Morgan/Reuters (1996). *RiskMetrics — Technical Document*, 4th ed. New York, 17 December 1996: §5.3.1.2 and Table 5.7 (effective days), §5.3.2.2 (the decay factor) and Table 5.9.
- Patton, A. J. (2011). "Volatility Forecast Comparison Using Imperfect Volatility Proxies." *Journal of Econometrics* 160 (1): 246–256. Working paper: Quantitative Finance Research Centre, University of Technology Sydney, Research Paper 175, April 2006, Section 3.
- Tsay, R. S. (2010). *Analysis of Financial Time Series*, 3rd ed. Hoboken, NJ: Wiley, chapter 3 (conditional heteroscedastic models).
- Zhang, L., P. A. Mykland, and Y. Aït-Sahalia (2005). "A Tale of Two Time Scales: Determining Integrated Volatility With Noisy High-Frequency Data." *Journal of the American Statistical Association* 100 (472): 1394–1411.
- arch 8.0.0: `arch_model`, `ARCHModel.fit` and `ARCHModelResult.forecast`.
- statsmodels 0.15.0: `het_arch` and `acorr_ljungbox`. Polars: `read_csv`.

- FinMind open data: `TaiwanStockTotalReturnIndex` (data_id TAIEX), retrieved 18 October 2026; `TaiwanVariousIndicators5Seconds` (the TAIEX every 5 to 60 seconds; TWSE the original source), retrieved 19 October 2026; `TaiwanStockPrice` (the TAIEX closes before each opening return), retrieved 11 and 18 October 2026; `TaiwanFuturesKBar` (TX, 2025) and `TaiwanFuturesDaily` (TX, the contract held), retrieved 19 October 2026.